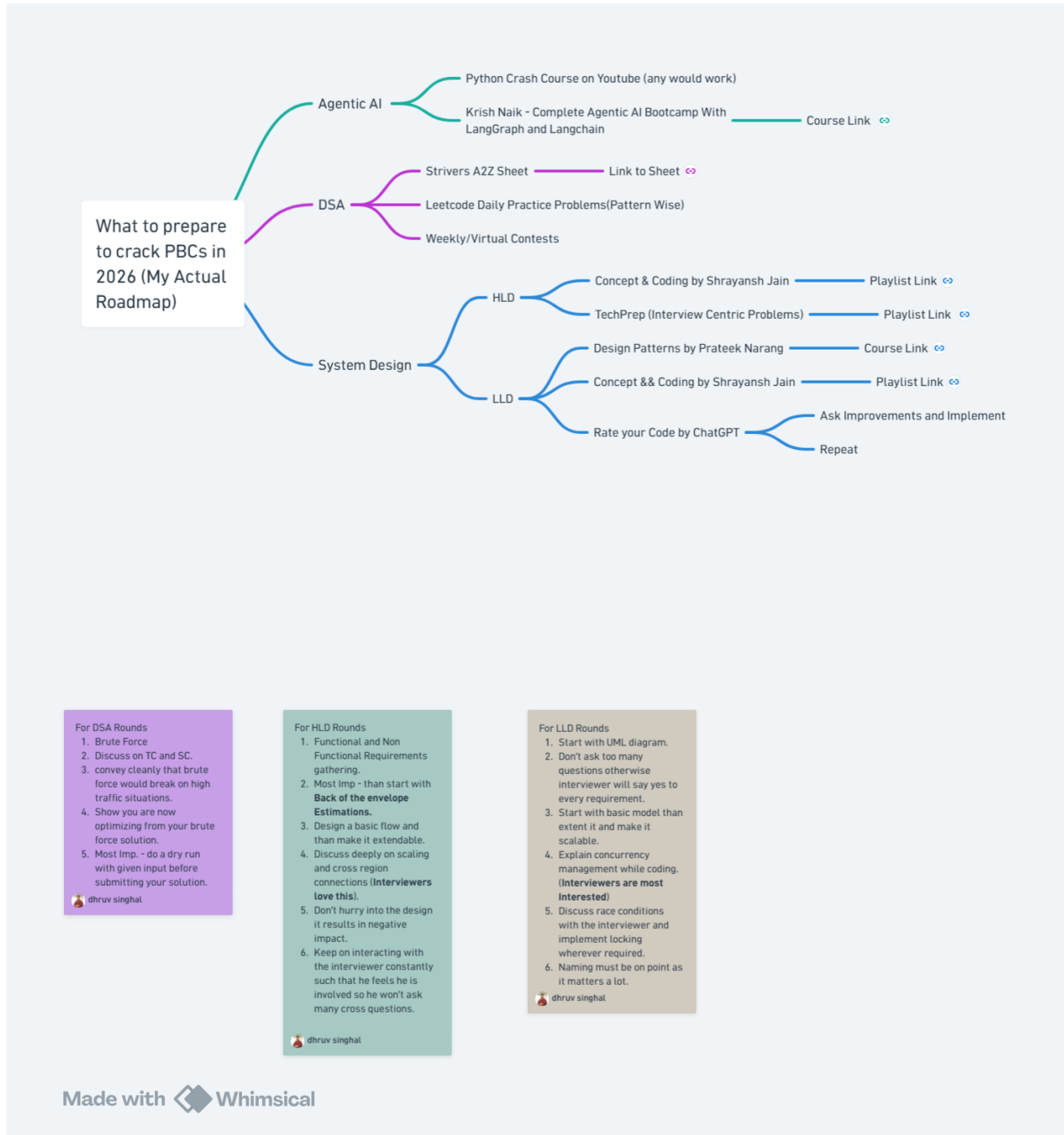


2026 Product Based Companies Preparation Roadmap

This roadmap is designed for software engineers preparing for Product Based Companies in 2026. The roadmap focuses on Agentic AI, DSA, and System Design with a strong emphasis on consistency, daily problem solving, and interview communication.



Preparation Timeline

Phase	Duration	Goal
Agentic AI	1.5 Months	Learn Python + Build AI Agents
DSA	3 Months	Complete Striver A2Z + Daily LeetCode
System Design	2 Months	Prepare HLD + LLD while continuing DSA

Phase 1 – Agentic AI (1.5 Months)

- Learn Python basics using any Python crash course available on YouTube.
- Understand Agentic AI workflows and architectures.
- Learn LangChain and LangGraph.
- Build AI Agents and mini automation projects.
- Deploy and experiment with practical use cases.

Resource:

Complete Agentic AI Bootcamp with LangGraph and LangChain:

<https://www.udemy.com/course/complete-agentic-ai-bootcamp-with-langgraph-and-langchain/>

Phase 2 – DSA (3 Months)

- Complete Striver's A2Z Sheet thoroughly.
- Solve problems pattern-wise.
- Practice Arrays, Strings, Binary Search, Trees, Graphs, DP, Greedy, and Backtracking.
- Participate in weekly or virtual contests.
- After completing the sheet, continue practicing 3–4 LeetCode questions daily.
- Focus on optimization and interview communication.

Resource:

Striver A2Z Sheet:

<https://takeuforward.org/dsa/strivers-a2z-sheet-learn-dsa-a-to-z>

Phase 3 – System Design (2 Months)

- Prepare both HLD and LLD.
- Continue doing DSA daily alongside System Design preparation.
- Learn scalability, caching, databases, queues, load balancing, and estimations.
- Learn SOLID principles, UML diagrams, Design Patterns, concurrency, and scalable code structure.
- Practice explaining tradeoffs and communication during interviews.

HLD Resources:

Shrayansh Jain Playlist:

<https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7>

TechPrep Playlist:

<https://www.youtube.com/playlist?list=PLRtLu6rCuAlkO-HiER3AKoKkSG5DPp9TX>

LLD Resources:

Prateek Narang Course:

<https://www.udemy.com/course/low-level-system-design/>

Shrayansh Jain Playlist:

https://www.youtube.com/playlist?list=PL6W8uoQQ2c61X_9e6Net0WdYZidm7zooW

My Interview Notes

For DSA Rounds

- 1 Brute Force
- 2 Discuss on TC and SC.
- 3 Convey clearly that brute force would break on high traffic situations.
- 4 Show how you are now optimizing from your brute force solution.
- 5 Most Important – do a dry run with given input before submitting your solution.

For HLD Rounds

- 1 Functional and Non Functional Requirements gathering.
- 2 Most Important – then start with Back of the envelope Estimations.
- 3 Design a basic flow and then make it extendable.
- 4 Discuss deeply on scaling and cross region connections (Interviewers love this).
- 5 Don't hurry into the design, it results in negative impact.
- 6 Keep interacting with the interviewer constantly such that he feels he is involved so he won't ask many cross questions.

For LLD Rounds

- 1 Start with UML diagram.
- 2 Don't ask too many questions otherwise interviewer will say yes to every requirement.
- 3 Start with basic model then extend it and make it scalable.
- 4 Explain concurrency management while coding (Interviewers are most interested).
- 5 Discuss race conditions with the interviewer and implement locking wherever required.
- 6 Naming must be on point as it matters a lot.

Consistency + Smart Preparation + Daily Execution = Results.